

SIMPLE STORAGE SERVICE (S3)

Class No: 5

S3 Definition:

Amazon S3 (Simple Storage Service) is a cloud-based object storage service from Amazon Web Services (AWS) that lets you store and retrieve virtually unlimited amounts of data with high scalability, security, and availability. It is widely used for backups, data lakes, application hosting, and analytics.

It is one of the important storage services. It is an object storage and independent in nature. So, we don't need any web server and we can host a static website using S3. It is a global service.

Buckets:

A container for storing objects. Names must be globally unique. In simple words, more like a folder in root level, which holds different files and folders. It stores unstructured data like videos, images and backups.

Key features:

- **Scalability:** Can handle data from gigabytes to exabytes without manual scaling.
- **Availability & Durability:** Designed for **99.99999999% (11 nines) durability**.
- **Security:** Offers encryption, fine-grained access controls, and compliance certifications.
- **Performance:** Optimized for fast retrieval and integration with AWS services.
- **Global Reach:** Buckets are region-specific, ensuring data stays within chosen regions unless moved.

Types of Buckets:

- **Public buckets:** can be accessed directly via internet like login page of the website.
- **Private buckets:** Can not be accessed directly via internet like transaction data, log files.

Permissions:

Permissions are used to define and change the public bucket to a private or vice-versa. There are two levels of permissions,

- Bucket level Permissions.
- Object level permissions.

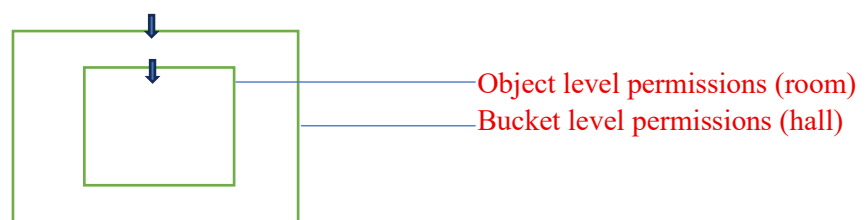


Fig 5.a Levels of permissions in bucket.

Both the paths must be unlocked to be a public bucket. If one of them is locked, then it is a private one.

ACL: Access Control List:

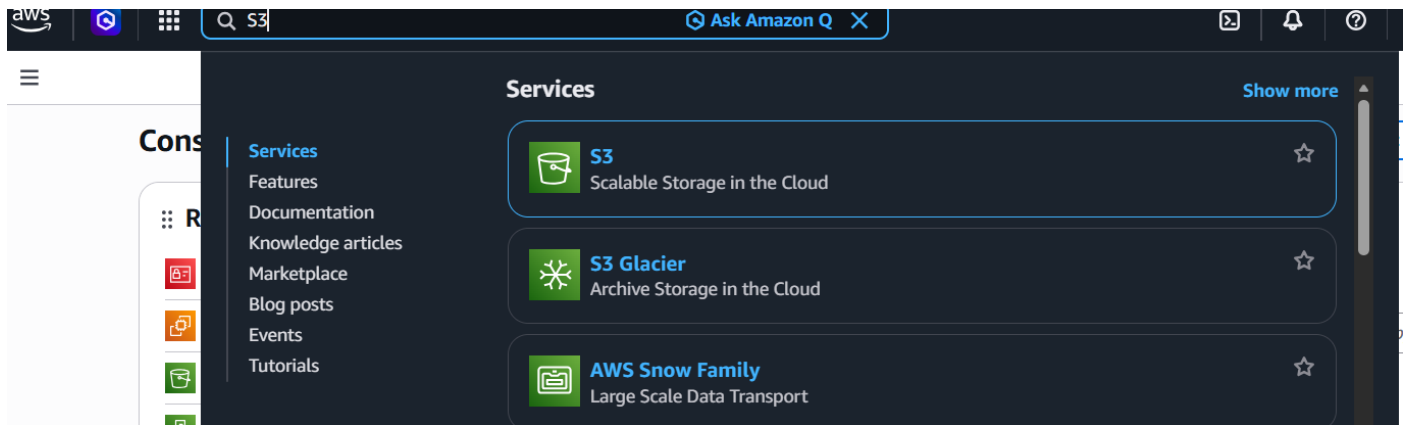
It determines the type of the bucket. If it is enabled - public bucket, disabled - private bucket.

Features of ACL:

- **Scope:** Can be applied at the *bucket* level or *object* level.
- **Granularity:** Permissions include actions like *READ*, *WRITE*, *READ_ACP* (read access control policy), and *WRITE_ACP*.
- **Use Case:** Often used for simple sharing scenarios, but AWS recommends using **bucket policies** or **IAM policies** for more fine-grained and scalable control.

Hands on: Creating a private bucket in S3 and changing it to public.

Step 1: Search S3 and click it.



Step 2: Choose general purpose, bucket names a global namespace. Hence it must be unique. Then disabled ACL under bucket ownership and block public access (as we are creating a private bucket).

Create bucket [Info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region

Europe (Stockholm) eu-north-1

Bucket type [Info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket namespace

Choose the namespace where you want to create your bucket. [Learn more](#)

☒ **Global namespace**
By default, S3 creates general purpose buckets in the global namespace.

☐ **Account Regional namespace (recommended)**
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name [Info](#)

Newprivatebucket06789

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional

Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Object Ownership [Info](#)

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

☒ **ACLs disabled (recommended)**
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☐ **ACLs enabled**
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership
Bucket owner enforced

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☒ **Block all public access**

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

- ☒ **Block public access to buckets and objects granted through new access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
- ☒ **Block public access to buckets and objects granted through any access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.
- ☒ **Block public access to buckets and objects granted through new public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
- ☒ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Also, if we want to add more security while sharing buckets to avoid deletion or rewriting them, we can enable object lock (Here I didn't enable it.) and click create bucket.

Bucket Versioning
Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning
☐ Disable
☒ Enable

Tags - optional
You can use bucket tags to analyze, manage and specify permissions for a bucket. [Learn more](#)

1 You can use s3:ListTagsForResource, s3:TagResource, and s3:UntagResource APIs to manage tags on S3 general purpose buckets for access control in addition to cost allocation and resource organization. To ensure a seamless transition, please provide permissions to s3:ListTagsForResource, s3:TagResource, and s3:UntagResource actions. [Learn more](#)

No tags associated with this bucket.

Add new tag

You can add up to 50 tags.

Default encryption [Info](#)
Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type [Info](#)
Secure your objects with two separate layers of encryption. For details on pricing, see [DSSE-KMS pricing](#) on the [Storage tab of the Amazon S3 pricing page](#).
☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)
☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)
☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)

Bucket Key
Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)
☐ Disable
☒ Enable

Advanced settings

Object Lock
Store objects using a write-once-read-many (WORM) model to help you prevent objects from being deleted or overwritten for a fixed amount of time or indefinitely. Object Lock works only in versioned buckets. [Learn more](#)
☒ Disable
☐ Enable
Permanently allows objects in this bucket to be locked. Additional Object Lock configuration is required in bucket details after bucket creation to protect objects in this bucket from being deleted or overwritten.

1 Object Lock works only in versioned buckets. Enabling Object Lock automatically enables Versioning.

1 After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.

Cancel

Create bucket

A private bucket is created. Now we are going to change it as a public one.

Step 3: Click on the created bucket to add a file. The file will be stored as an object. Click on the upload to add a file.

newprivatebucket06789 [Info](#)

Objects

Metadata

Properties

Permissions

Metrics

Management

Access Points

Objects (0)
Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

☒ Show versions

< 1 > [Settings](#)

☐	Name	Type	Last modified	Size	Storage class
No objects You don't have any objects in this bucket.					

Upload

Step 4: Now click add file and click upload.

Upload [Info](#)
Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose [Add files](#) or [Add folder](#).

Files and folders (1 total, 75.8 KB)
All files and folders in this table will be uploaded.

☐	Name	Type	Size
☐	amber_girl.jpg	Image/jpeg	75.8 KB

Remove Add files Add folder

Destination [Info](#)
Destination
[s3://newprivatebucket06789](#)

Destination details
Bucket settings that impact new objects stored in the specified destination.

Permissions
Grant public access and access to other AWS accounts.

Properties
Specify storage class, encryption settings, tags, and more.

Cancel

Upload

Step 5: Click on the object, we can see the object url has been created. Copy and paste it in browser. It will show access denied. But we see them inside the console by clicking the open option on the object page top right side.

amber_girl.jpg Info

[Copy S3 URI](#) [Download](#) [Open](#) [Object actions](#)

[Properties](#) [Permissions](#) [Versions](#)

Object overview

Owner
0664ff8b5d2fc7fe499721e9d7d5488c856855e61f322c1c4fa31c5bcc89717e

AWS Region
Europe (Stockholm) eu-north-1

Last modified
March 13, 2026, 19:06:49 (UTC+05:30)

Size
75.8 KB

Type
jpg

Key
amber_girl.jpg

S3 URI
[s3://newprivatebucket06789/amber_girl.jpg](https://newprivatebucket06789/amber_girl.jpg)

Amazon Resource Name (ARN)
arn:aws:s3::newprivatebucket06789/amber_girl.jpg

Entity tag (Etag)
[7d9127bf5d518f907c04f50fa6ba889e](#)

Object URL
https://newprivatebucket06789.s3.eu-north-1.amazonaws.com/amber_girl.jpg

Step 6: To convert this into public, first we need to enable the bucket level permissions.

Go to the bucket we have created and click on it, **choose permissions**.

newprivatebucket06789 Info

[Objects](#) [Metadata](#) [Properties](#) [Permissions](#) [Metrics](#) [Management](#) [Access Points](#)

Permissions overview

Access finding
Access findings are provided by IAM external access analyzers. Learn more about [How IAM analyzer findings work](#).
[View analyzer for eu-north-1](#)

Block public access (bucket settings) [Edit](#)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

Block all public access
☒ On
► Individual Block Public Access settings for this bucket

Bucket policy [Edit](#) [Delete](#)

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

Public access is blocked because Block Public Access settings are turned on for this bucket
To determine which settings are turned on, check your Block Public Access settings for this bucket. Learn more about [using Amazon S3 Block Public Access](#)

Now, Choose edit in the **Block public access** disable it and save changes.

Edit Block public access (bucket settings)

Edit Block public access (bucket settings) Info

Block public access (bucket settings)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☐ **Block all public access**
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- ☐ **Block public access to buckets and objects granted through any access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.
- ☐ **Block public access to buckets and objects granted through new public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
- ☐ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

[Cancel](#) [Save changes](#)

On the same bucket page, scroll down choose edit in the object ownership. **Enable ACL** to convert it as a public bucket. Then acknowledge and save changes.

Edit Object Ownership [Info](#)

Object Ownership

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

☐ ACLs disabled (recommended)
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☒ ACLs enabled
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

⚠ We recommend disabling ACLs, unless you need to control access for each object individually or to have the object writer own the data they upload. Using a bucket policy instead of ACLs to share data with users outside of your account simplifies permissions management and auditing.

⚠ Enabling ACLs turns off the bucket owner enforced setting for Object Ownership
Once the bucket owner enforced setting is turned off, access control lists (ACLs) and their associated permissions are restored. Access to objects that you do not own will be based on ACLs and not the bucket policy.
☒ I acknowledge that ACLs will be restored.

Object Ownership

☒ Bucket owner preferred
If new objects written to this bucket specify the bucket-owner-full-control canned ACL, they are owned by the bucket owner. Otherwise, they are owned by the object writer.
☐ Object writer
The object writer remains the object owner.

ⓘ If you want to enforce object ownership for new objects only, your bucket policy must specify that the bucket-owner-full-control canned ACL is required for object uploads. [Learn more](#)

[Cancel](#)

[Save changes](#)

Scroll down we can see the **ACL control list**, choose edit and enable Read and write for everyone. Acknowledge and save changes.

Edit access control list (ACL) [Info](#)

Access control list (ACL)

Grant basic read/write permissions to other AWS accounts. [Learn more](#)

Grantee	Objects	Bucket ACL
Bucket owner (your AWS account) Canonical ID: 0664f8b5d2fc7fe499721e9d7d5488c856855e61f322c1c4fa31c5bcc89717e	☒ List ☒ Write	☒ Read ☒ Write
Everyone (public access) Group: http://acs.amazonaws.com/groups/global/AllUsers	☒ List ☒ Write	☒ Read ☒ Write
Authenticated users group (anyone with an AWS account) Group: http://acs.amazonaws.com/groups/global/AuthenticatedUsers	☐ List ☐ Write	☐ Read ☐ Write
S3 log delivery group Group: http://acs.amazonaws.com/groups/s3/LogDelivery	☐ List ☐ Write	☐ Read ☐ Write

⚠ When you grant access to the Everyone or Authenticated users group grantees, anyone in the world can access the objects in this bucket.
[Learn more](#)
☒ I understand the effects of these changes on my objects and buckets.

Access for other AWS accounts

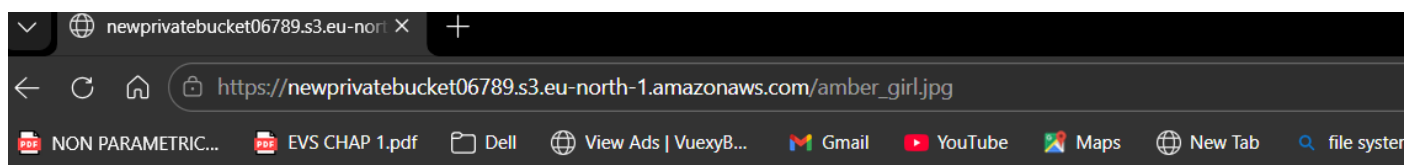
No other AWS accounts associated with the resource.

[Add grantee](#)

[Cancel](#)

[Save changes](#)

As we have only opened the bucket level permissions, still we can not access the object from outside via internet in a browser. Still, it will show access denied. Now we have to open and set up the object level permissions.



This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
<?xml version="1.0" encoding="UTF-8" ?>
<Error>
  <Code>AccessDenied</Code>
  <Message>Access Denied</Message>
  <RequestId>AG2MH4J9Z3KNV440</RequestId>
  <HostId>4Kiakuw9DhSpICwywLda+2SHgF/Rctrww0bwrOGIdSDtMMQEZq1Pi0BK0hrajCmxcdfyzINBmca01D7JXNABBB6wXtKxmbt3</HostId>
</Error>
```

Step 7: Open the bucket, click the file we have added (It will be under the object) and click permissions section.

amber_girl.jpg Info [Copy S3 URI](#) [Download](#) [Open](#) [Object actions](#)

Properties **Permissions** Versions

Access control list (ACL) [Edit](#)

Grant basic read/write permissions to AWS accounts. [Learn more](#)

Grantee	Object	Object ACL
Object owner (your AWS account) Canonical ID: 0664ff8b5d2fc7fe499721e9d7d5488c856855e61f322c1c4fa31c5bcc89717e	Read	Read, Write
Everyone (public access) Group: http://acs.amazonaws.com/groups/global/AllUsers	-	-
Authenticated users group (anyone with an AWS account) Group: http://acs.amazonaws.com/groups/global/AuthenticatedUsers	-	-

Now click the **edit**, read access to everyone for objects& objects ACL. Acknowledge and save changes.

[newprivatebucket06789](#) > [amber_girl.jpg](#) > Edit access control list

Edit access control list Info

Access control list (ACL)
Grant basic read/write permissions to AWS accounts. [Learn more](#)

Grantee	Objects	Object ACL
Object owner (your AWS account) Canonical ID: 0664ff8b5d2fc7fe499721e9d7d5488c856855e61f322c1c4fa31c5bcc89717e	☒ Read	☒ Read ☒ Write
Everyone (public access) Group: http://acs.amazonaws.com/groups/global/AllUsers	☒ ☒ Read	☒ ☒ Read ☐ Write
Authenticated users group (anyone with an AWS account) Group: http://acs.amazonaws.com/groups/global/AuthenticatedUsers	☐ Read	☐ Read ☐ Write

When you grant access to the Everyone or Authenticated users group grantees, anyone in the world can access this object.
[Learn more](#)

☒ I understand the effects of these changes on this object.

Access for other AWS accounts
No other AWS accounts associated with the resource.

[Add grantee](#)

Specified objects

Name	Type	Version ID	Last modified	Size
amber_girl.jpg	jpg	-	March 13, 2026, 19:06:49 (UTC+05:30)	75.8 KB

[Cancel](#) [Save changes](#)

Now we can access the file(object) via internet from a browser (outside) using the object URL and from the S3 console(inside).

Note: in order to delete the bucket, first empty its objects using the empty button. Then choose the delete option to delete the bucket.

Buckets

General purpose buckets

All AWS Regions

Directory buckets

General purpose buckets (1/1) Info

Buckets are containers for data stored in S3.

< 1 >

Name	AWS Region	Creation date
newprivatebucket06789	Europe (Stockholm) eu-north-1	March 13, 2026, 19:02:59 (UTC+05:30)